NAYEM SIDDIKI

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GitHub: https://github.com/NayemSiddiki02

EDUCATION

American International University-Bangladesh (AIUB) Expected graduation in Fall 2026 - 2027
Bachelor of Science in Computer Science & Engineering (CSE)
CGPA: 3.96/4.00
Relevant Coursework:

- Data Structures and Algorithms (DSA)
- Natural Language Processing (NLP)
- Object-Oriented Programming (OOP)
- Machine Learning (ML)
- Numerical Methods

Dhaka College 2019 - 2022
Higher Secondary Certificate (Science)
GPA: 5.00/5.00

Chak-kirti High School, Nawabganj 2014 - 2019
Secondary School Certificate (Science)
GPA: 5.00/5.00

RESEARCH INTERESTS

- Machine Learning (ML)
- Deep Learning (DL)
- Natural Language Processing (NLP)
- Computer Vision & Pattern Recognition (CVPR)
- Transformer-based Language Models
- Physics Informed Neural Network (PINN)
- Fluid Mechanics
- Quantum Droplets
- Quantum Computing

EXPERIENCE

Research Intern
ELITE Research Lab LLC June 2026 – Present

- Gaining Research experience with project and research article writing.

ACADEMIC PROJECTS

Khoja-Khuji: A lost and found management system webapp

- A lost and found management system based on web application
- Build with HTML, CSS, JavaScript, PHP and mySQL
- Applied useful feature to build this project

AI-Generated Text Detection using BERT and DistilBERT 🔗

- Fine-tuned transformer-based models for binary text classification.
- Implemented preprocessing, tokenization, and model evaluation pipelines.
- Compared performance of BERT and DistilBERT architectures.

- Conducted experimental evaluation using accuracy, precision, and F1-score metrics.

Explainable Lightweight Deep Learning for Common Skin Disease Classification in Low-Resource Clinical Environments

- A lightweight skin disease classification model based on MobileNetV3 was developed for efficient and accurate prediction after comparing with other 2 model.
- Transfer learning concept was applied, and the final classification layer was modified to adapt the model to the target data-set.
- Grad-CAM was integrated to provide visual explanations of the model predictions to improve interpretability and clinical trust.

An Explainable Multi-Task Approach to Vehicle Classification and Localization on Bangladeshi Roads Using VGG16 and ResNet18

- An end-to-end multi-task deep learning model to perform vehicle classification and localization using Bangladeshi road vehicle dataset named Sorokh-Poth dataset.
- Used CNN based pretrained model (VGG16 and ResNet18) as base model.
- Used GradCAM for visualization and increase trust.

Fake News Detection using Naive Bayes 🔗

- Applied TF-IDF feature extraction for text vectorization.
- Implemented probabilistic classification using Naive Bayes.
- Evaluated model performance using standard classification metrics.

Computer Graphics: Village View 🔗

- Developed graphical simulation using C++ and OpenGL.
- Implemented rendering techniques, geometric transformations, and animation logic.
- Applied computer graphics pipeline concepts in scene construction.

Hostel Management System (C#) 🔗

- Designed and implemented a role-based management system.
- Applied object-oriented design principles and modular architecture.

Bus Ticket Management System (Java) 🔗

- Developed a structured ticket management application.
- Applied OOP concepts and system modularization.

Enhancement on Pathao App 🔗

- Suggested some enhancements on Pathao App.
- Proposed all the planning for that.

Arduino based Smart Mop/Brush Cleaner Robot 🔗

- Developed an autonomous 4-wheel drive robot using an Arduino Uno and L293D shield to automate domestic cleaning tasks.
- Integrated an HC-SR04 ultrasonic sensor on a servo motor for 180º obstacle scanning and real-time path planning.
- Programmed a "Backward-then-Soft-Left" algorithm with averaged distance sensing to filter noise and ensure smooth navigation.

Arduino-Based Fire & Smoke Detection System 🔗

- Designed an automated safety system using Arduino Uno to provide real-time audible and visual alerts for hazardous conditions.
- Integrated infrared flame sensors and MQ-2 gas sensors to accurately detect fire and flammable smoke concentrations.
- Engineered an I2C-based alert interface using an active buzzer and a 16×2 LCD module to display system status and warning messages.

- Developed embedded C++ logic with customized threshold values and active-low signal processing for reliable environmental monitoring.

Rain Detection and Alarming Using 555 Timer IC.

- Engineered a Rain Detection System using a 555 Timer IC in astable multivibrator mode to trigger an automated audible alarm.
- Implemented a BC547 transistor switching circuit to interface with a rain sensor, allowing for precise control of the timer's RESET pin.
- Optimized signal output by calculating and configuring external resistor and capacitor values to achieve a consistent 0.72Hz alert frequency.

Design and implementation of a DC mobile charger with fast charging.

- Engineered a high-efficiency power supply utilizing a DC Buck-Boost module to provide adjustable voltage and current regulation.
- Implemented pulse-based switching control using an LM555 Timer and complementary transistors to optimize power delivery.
- Constructed a full-wave bridge rectifier circuit with a high-capacity smoothing capacitor to convert AC mains to stable DC.

ACADEMIC HONORS AND AWARDS

Dean's List Honors

American International University–Bangladesh

- Spring 2023-2024 (CGPA: 3.99)
- Fall 2024-2025 (CGPA: 3.98)

Certificate of Participation

16th National Undergraduate Mathematics Olympiad (2025)
Bangladesh Mathematical Society - AF Mujibur Rahman Foundation

TECHNICAL SKILLS

Programming Languages:

C, C++, Java, C#, Python, MATLAB, Assembly Language, MySQL, Microsoft MySQL, HTML, CSS, JavaScript, CSS

Tools & Platforms:

MATLAB, Jupyter Notebook, Multisim, Arduino IDE, Proteus, AutoCAD 2007, Cisco Packet tracer, XAMP

Core Areas:

Data Structures, Algorithms, Object-Oriented Programming, Machine Learning, Deep Learning, Natural Language Processing

CERTIFICATIONS

- Foundational C# with Microsoft - freeCodeCamp (2025)
- MATLAB Onramp - MathWorks (2025)
- IT Essentials: PC Hardware and Software - Cisco Networking Academy (2023)
- Getting Started with Cisco Packet Tracer - Cisco Networking Academy (2025)
- Microsoft Word Basics - MuktoPaath (2020)
- Introduction to Gitn - datacamp (2026)

LANGUAGES

Bengali - Native Language
English - Limited Working Proficiency
Arabic - Elementary Proficiency

VOLUNTEERING

Management Volunteer - Bangladesh Mathematical Olympiad (BdMO) 2025

- Provided voluntary service as an exam invigilator.

CONTACT

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REFERENCES

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